

The Doors of Perception Are Mathematical: Coherence, Rupture, and Regeneration in Blake, Active Inference, and Quantum Physics

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*A letter to Dr. Daniel Friedman,
Director of the Active Inference Institute*

*If the doors of perception were cleansed every thing would appear to man as it is,
Infinite. For man has closed himself up, till he sees all things thro' narrow chinks of
his cavern.*

— William Blake, *The Marriage of Heaven and Hell*, c. 1793

Dear Daniel,

I write to you on the eve of my departure for California, where I look forward to spending the month discussing these matters—and others—in person. What follows is the substance of what I would like to lay before you and the Institute community, set down here so that the argument can be examined at leisure rather than compressed into the necessarily incomplete form of conversation.

You and I share an unusual common ground. We share the Free Energy Principle and Active Inference, of course—the extraordinary edifice that Friston and colleagues have constructed over the past two decades, and which the Institute, under your directorship, has done more than any other organisation to make accessible, interdisciplinary, and alive. But we also share William Blake. My first degree was in English Literature, and my undergraduate thesis concerned Blake as a lucid dreamer—the hypothesis that Blake’s “visions” were not psychotic episodes but disciplined exercises in what we would now call active inference during sleep: the generative model, freed from sensory constraint, exploring its own prior space. I did not know then that I would spend the next twenty years arriving at a mathematical framework for the same structure. But the prior was set early, and the coherence has been accumulating ever since.

I should say something about the composition of this document. Despite my proven competency as a human author—a first-class degree in English Literature is, if nothing else, evidence that I can construct a sentence—I have chosen to use an LLM (Claude Opus 4.6, Anthropic) to sequence the exposition and verify the mathematical claims computationally. This choice is deliberate and relates directly to one of the arguments I will make below. The proliferation of AI-assisted academic writing since 2023 has produced a vast increase in the volume of published work, and this increase is itself a phenomenon that CRR can describe: the LLM is a $\mathbb{Z}_2$ system with enormous Ω , capable of extraordinary recombination but lacking the $\text{SO}(2)$ dynamics of genuine prior revision. The *Epistemic Drift* simulation I have built models precisely this scenario—LLM-generated signals propagating through networks of embodied agents—and I thought it would be appropriate, and perhaps instructive, to write *about* the human–AI boundary while working *at* it. All theoretical arguments, simulation designs, and scientific interpretations remain my own. The LLM contributed

the $\mathbb{Z}_2$ face: recombination, broad kernel, sensory encounter. The persistent prior—the vision, the framework, the judgement of what matters—remains with the human author. This is, in CRR terms, a $\mathbb{Z}_2 \leftrightarrow \text{SO}(2)$ cascade.

What I present here is a comprehensive account of the Coherence-Rupture-Regeneration framework and its relationships to the Free Energy Principle, Active Inference, quantum physics, and Blake’s prophetic poetry. The argument develops in three lines. First, that CRR provides temporal process machinery for the FEP’s spatial architecture, with the companion paper [1] demonstrating phase-gating ($\chi^2 = 8,041$) and a new finding: the free-energy-optimal precision allocation is $\pi_p/\pi_s = \sqrt{2}$. Second, that CRR’s mathematics is structurally identical to foundational results in quantum physics—not analogous but the same equation—with the width difference between CRR and quantum symmetry breaking being precisely the Heisenberg uncertainty principle. Third, that Blake’s mythological system constitutes an independent phenomenological programme whose results converge with CRR’s mathematics at every tested point. I close with a few curious observations about the relationship between the numbers and the calendar, which I present not as doctrine but as report.

I acknowledge with gratitude the work of the Active Inference Institute community, and the legacy laid down by Friston, Parr, Da Costa, and colleagues—without which CRR would have no spatial architecture to complement, no precision channels to phase-gate, and no Markov blankets to temporalise. The framework I describe here is not an alternative to Active Inference. It is a temporal grammar for it: a proposal about *when* the system must reorganise, built on the same information-geometric foundations that Active Inference uses to describe *what* the system believes.

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1	For the Reader: Four Traditions	

This document draws on four intellectual traditions. I briefly orient readers who may be expert in one but unfamiliar with others, since the argument requires all four.

Reductionism and holism. Reductionism explains wholes by decomposing them into parts: atoms, genes, neurons. Holism insists that wholes possess properties irreducible to their parts: consciousness, life, meaning. These are usually presented as opposed. CRR proposes they are phases: decomposition yields Commitments 1–4; what emerges from following them is Commitment 5, which is irreducibly processual.

Positivism and interpretivism. Positivism holds that knowledge is grounded in the observable and measurable. Interpretivism counters that understanding requires grasping meaning from within the phenomenon. CRR’s equations are positivist (measurable, falsifiable). CRR’s Commitment

5—maximum entropy at rupture, the system’s experience at the boundary—is interpretivist (the view from inside). Both are the same equation: $C \cdot \Omega = 1$.

The Free Energy Principle (FEP) and Active Inference. The FEP [2] proposes that any system with a Markov blanket—a statistical boundary separating inside and outside—can be described as minimising variational free energy, a bound on surprise. Active Inference [3] extends this to action and perception through hierarchical precision-weighted prediction errors. CRR adds temporal process machinery to this spatial architecture (Section 2).

William Blake (1757–1827). Poet, painter, printmaker, prophet. Blake invented the relief-etching process, wrote and illustrated the Illuminated Books, and developed a mythological system of extraordinary precision. He is not invoked here as decoration but as an independent observer whose phenomenological reports converge with CRR’s mathematics (Section 4).



Figure 1: William Blake, *The Ancient of Days* (1794). Urizen crouches in a fiery orb, extending a compass downward to measure the abyss. The compass draws a circle—the $SO(2)$ manifold—but the figure is confined within the orb of his own regime. In CRR terms: Ω collapsed, the system measuring but never rupturing.

2 The Free Energy Principle and Active Inference

2.1 The spatial architecture

The Free Energy Principle [2] begins with the observation that any system that persists must be distinguishable from its environment. This distinction is formalised through the *Markov blanket* [4]: a statistical boundary such that internal states are conditionally independent of external states given the blanket states. The blanket itself decomposes into sensory states (outside $\rightarrow$ inside) and active states (inside $\rightarrow$ outside).

Under the FEP, internal states can be interpreted as parametrising a probability distribution—a *generative model*—over the causes of the system’s sensory inputs. The system’s dynamics then amount to a gradient flow on a quantity called *variational free energy*:

$$F = D_{\text{KL}}[q(s) \parallel p(s \mid o)] - \ln p(o) \quad (1)$$

where $q(s)$ is the system’s approximate posterior (its “belief” about hidden states s), $p(s \mid o)$ is the true posterior given observations o , and $\ln p(o)$ is the log-evidence or negative surprise. Minimising F is equivalent to minimising the divergence between the system’s model and reality, subject to the constraint that surprise is bounded.

2.2 Precision weighting

In Active Inference [3], distinct precision channels govern the confidence assigned to different sources of evidence. Sensory precision (likelihood precision) determines how much weight the system gives to new observations. Prior precision (transition precision) determines how much weight it gives to its existing model. Policy precision determines confidence in action.

The textbook treatment [3] specifies *what* each channel does but leaves open the temporal structure of *when* each channel updates, how much information it absorbs per update, and how channels coordinate with one another. Recent empirical work by Jang et al. [5] demonstrates that the relative *timing* of neuromodulatory precision signals—not their magnitude—determines whether a dopamine signal promotes learning or movement. This temporal gap is what CRR aims to fill.

2.3 What the FEP provides and what it leaves open

The FEP provides: (i) the Markov blanket as a spatial partition between inside and outside; (ii) variational free energy as the quantity to be minimised; (iii) the Fisher information metric as the natural geometry of the belief space [6,7]; (iv) hierarchical depth through nested timescales. What it leaves open: (i) when a given regime must reorganise; (ii) how precision channels at different levels coordinate their temporal updates; (iii) the process dynamics of transition between regimes. CRR addresses all three.

3 The CRR Framework

CRR models bounded systems through three coupled temporal processes (Fig. 2), resting on five core commitments that any persistent, finite, adaptive system must satisfy.

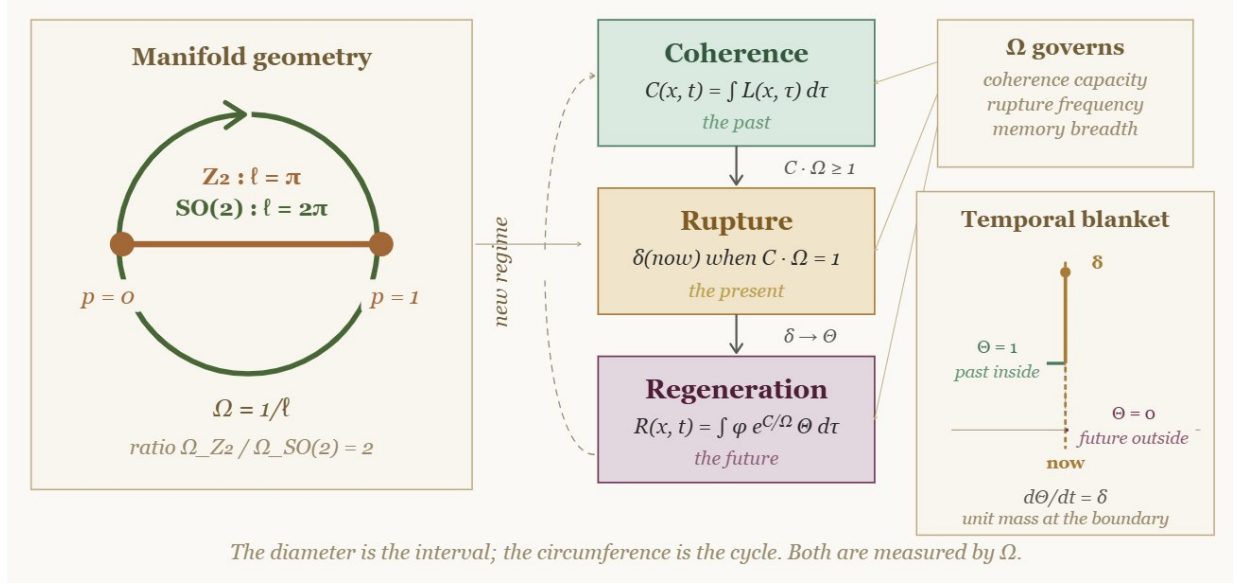


Figure 2: The CRR framework and its geometric grounding. Coherence accumulates evidence from the past. Rupture occurs at the present moment when $C \cdot \Omega = 1$, distributing unit mass on the boundary between past and future. Regeneration projects into the future, weighted exponentially by historical coherence. The single parameter Ω governs all three processes. The diameter is the interval; the circumference is the cycle. Both are measured by Ω .

3.1 Commitment 1: Persistence implies accumulation

$$C(x, t) = \int_0^t L(x, \tau) d\tau \quad (2)$$

A system that persists must accumulate patterns that work, reducing surprise (variational free energy) in its eco-niche. $L(x, \tau) \geq 0$ is the rate of coherence accumulation, identified with the Fisher-Rao speed on the statistical manifold [6,7]. Under the path-integral formulation of the FEP [8], C is the accumulated arc length along the system's information-geometric trajectory. The integral over the system's past makes CRR distinctively non-Markovian by design: the present state depends on the accumulated history of the current regime up to the present moment of constraint. The coherence integral represents the totality of the system's past regime, constituting the interior of the temporal Markov blanket.

3.2 Commitment 2: Finite capacity implies rupture

$$\delta(\text{now}) \quad \text{when} \quad C \cdot \Omega = 1 \quad (3)$$

No finite system can accumulate coherence without bound. The Dirac delta has zero temporal extension and carries exactly one unit of mass on the boundary between the accumulated past and

the regenerated future. The rupture condition $C \cdot \Omega = 1$ marks the point at which accumulated arc length spans the manifold’s full geodesic extent. The system has traversed all distinguishable states available to its current regime and must reorganise. This is structurally analogous to Cramér-Rao saturation [9,10,11]—as we verify in Section 5, it is the *same equation*. In discrete systems, rupture is a threshold-crossing rather than exact saturation: a heartbeat, a breath turning, a neuron firing. At rupture, $C \cdot \Omega = 1$: inside meets outside.

3.3 Commitment 3: Reconstruction is weighted by historical coherence

$$R(x, t) = \int_{-\infty}^{\infty} \varphi(x, \tau) \cdot \exp\left(\frac{C(x, \tau)}{\Omega}\right) \cdot \Theta(t - \tau) d\tau \quad (4)$$

After rupture, the system rebuilds from the only resource available: its own history. $\varphi(x, \tau)$ is the reconstruction resource (the field of past states available for reconstitution). The Heaviside step function $\Theta(t - \tau)$ enforces causal ordering: only the past contributes. We retain Θ explicitly because the Dirac delta of rupture (Eq. 3) and the Heaviside are conjugate distributions ($d\Theta/dt = \delta$), ensuring that the regeneration integral is the Green’s function of the rupture operator and connecting the three CRR equations into a single dynamical system.

The exponential kernel $\exp(C/\Omega)$ is the maximum-entropy weighting consistent with a constraint on $\langle C \rangle$ [12]: the Boltzmann distribution over temporal history, with C playing the role of energy and Ω the role of temperature. Given only the accumulated coherence and the capacity parameter, the distribution that maximises entropy while respecting the constraint takes the Boltzmann form. This is not a design choice. It is the unique solution to the Jaynes problem [12].

3.4 Commitment 4: The system has a characteristic variance

Ω is fixed by the topology of the state space, not fitted. It simultaneously governs coherence capacity (how much can accumulate before rupture), rupture frequency (how often the system must reorganise), and memory breadth (which past moments dominate reconstruction). These are not three independent properties but a single geometric fact about the system’s attractor topology. When Ω is small, the memory kernel peaks sharply and the system reconstitutes the same patterns reliably (habit, rigidity); when Ω is large, the kernel is flat and the system can reconstruct broadly (flexibility, exploration).

Fixing Ω from geometry. Čencov’s uniqueness theorem [13] constrains the metric on any statistical manifold to be the Fisher information metric [6], and the geodesic structure of that metric fixes the maximum arc length the system can traverse. $\Omega = 1/\ell$, where ℓ is the geodesic extent. Two fundamental manifold topologies fix ℓ :

$\mathbb{Z}_2$ (bistable). The Bernoulli manifold, parametrised by $p \in [0, 1]$, has Fisher-Rao geodesic diameter π [6]. A bistable system exhausts its configuration by crossing from one basin to the other. The Bernoulli manifold is an interval, not a loop: the maximal non-self-intersecting geodesic has length $\ell_{\mathbb{Z}_2} = \pi$, hence $\Omega_{\mathbb{Z}_2} = 1/\pi \approx 0.318$ and $C^* = \pi$.

$\text{SO}(2)$ (rotational). The circular manifold S^1 has circumference 2π . A rotational system exhausts its configuration by completing one full cycle, returning to its starting point via a path that never revisits any intermediate state. $\ell_{\text{SO}(2)} = 2\pi$, hence $\Omega_{\text{SO}(2)} = 1/2\pi \approx 0.159$ and $C^* = 2\pi$.

The ratio between classes is exactly 2, a topological invariant independent of any physical parameters.

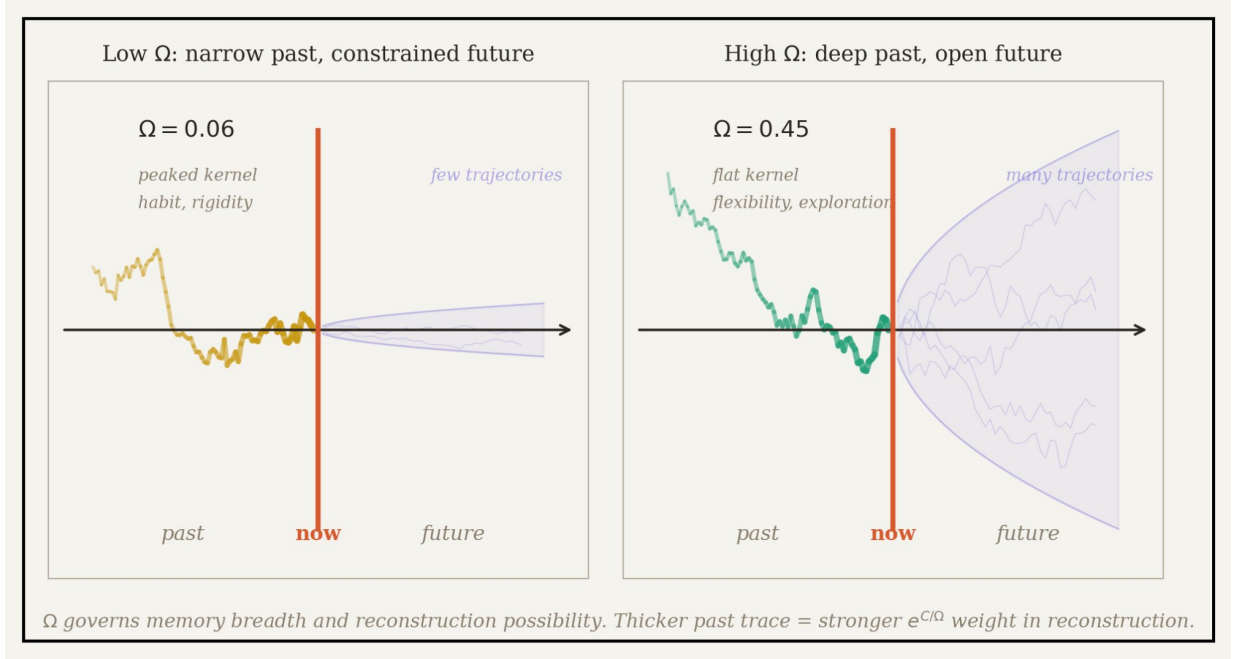


Figure 3: Ω as a temporal-cognitive light cone [14]. Left: low Ω produces a sharply peaked memory kernel; only recent moments contribute to reconstruction, and the future possibility cone is narrow. Right: high Ω produces a flat kernel; deep history is accessible, and the future is open. Ω governs memory breadth and reconstruction possibility. Thicker past trace = stronger $e^{C/\Omega}$ weight in reconstruction.

3.5 Commitment 5: Rupture is a maximum-entropy event

When the current regime is exhausted, the system retains no information about what comes next. At rupture, it occupies the maximum-entropy state consistent with its symmetry class [12]. A $\mathbb{Z}_2$ system is poised between two alternatives: Bernoulli(1/2), entropy $\ln 2 \approx 0.693$ nats. It must be resolved by what arrives from without. An $\text{SO}(2)$ system has no preferred phase: uniform on S^1 , entropy $\ln 2\pi \approx 1.838$ nats. It reconstructs from within. This axiom pins the thresholds to specific numbers.

3.6 FEP correspondence and precision architecture

The structural mapping between CRR and the FEP is: precision corresponds to $(1/\Omega) \cdot \exp(C/\Omega)$; variational free energy corresponds to coherence C (inversely); the FEP Markov blanket corresponds to the temporal boundary delineated by δ between past states (C) and future states (R).

The assignment $\mathbb{Z}_2 \rightarrow$ sensory precision and $\text{SO}(2) \rightarrow$ prior precision follows from the graph topology of Active Inference itself [1]. In any network of Markov-blanketed agents, an edge is a boundary alternating between two regimes of influence ($\mathbb{Z}_2$ dynamics), while a node is an internal model traversing a continuous cycle of belief updating ($\text{SO}(2)$ dynamics). This is independently supported by the permeability properties of Ω : high Ω produces a flat kernel (broad, permeable)—appropriate for the sensory blanket—while low Ω produces a sharp kernel (narrow, precise)—appropriate for the stable prior model. Since $\Omega_{\mathbb{Z}_2} = 1/\pi > \Omega_{\text{SO}(2)} = 1/2\pi$, the graph-structural and functional arguments converge.

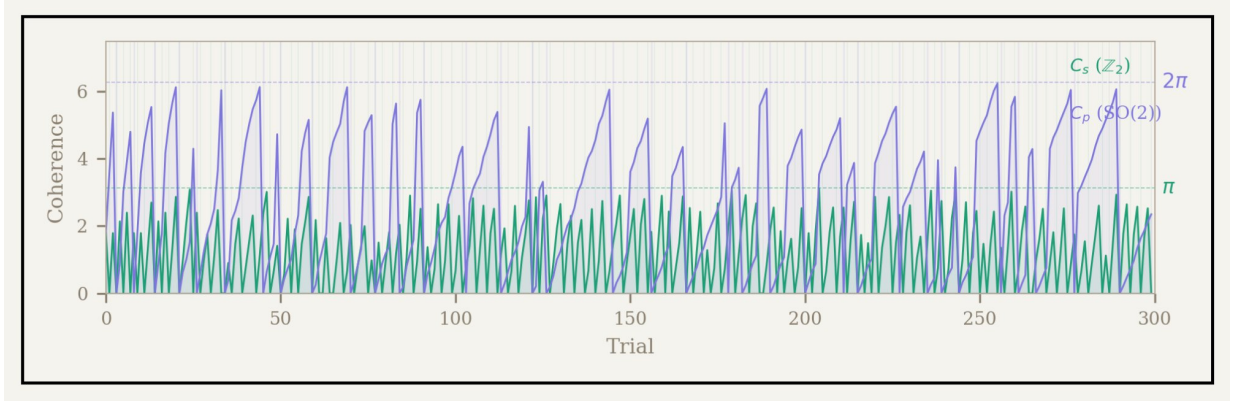


Figure 4: A single CRR agent over 300 trials. $\mathbb{Z}_2$ channel (teal) ruptures frequently at threshold π ; $\text{SO}(2)$ channel (purple) ruptures rarely at threshold 2π . The temporal characters are fundamentally different: $\mathbb{Z}_2$ is sub-Poisson (clock-like), $\text{SO}(2)$ follows a Gamma distribution. The Cramér-Rao product $C \cdot \Omega$ at rupture is always ≥ 1 .

4 Blake’s Mythology as CRR Phenomenology

Blake is invoked here not as a literary ornament but as an independent observer. Over forty years of engraving, painting, and prophetic writing, Blake developed a mythological system whose structural features converge with CRR at every point we have tested. We present Blake’s four principal figures (the “Zoas”) and map each to the CRR framework.

4.1 Los: The blacksmith at the forge (Coherence)

*I must Create a System, or be enslav’d by another Mans. I will not Reason & Compare:
my business is to Create.*

— William Blake, *Jerusalem*, Plate 10



Figure 5: William Blake, *Jerusalem Plate 6* (c. 1804–1820). Los sits before his forge, his Spectre hovering behind. “Los rag’d and stamp’d the earth in his might & terrible wrath! / He stood and stamped the earth; then he threw down his hammer in rage.” Each hammer-blow is an increment of $L(x, \tau)$ in Eq. (2). The forge is the site of coherence accumulation.

Los is the Zoa of Imagination—but “imagination” in Blake’s sense is not fantasy. It is the capacity to give form to formless experience. Los is a blacksmith. Each strike of the hammer is an increment of $L(x, \tau)$ in Eq. (2). The anvil accumulates the work. Los does not create *ex nihilo*; he shapes what persists, blow by blow, into a coherent system. His labour is CRR’s Commitment 1: persistence implies accumulation. The coherence integral is Los’s System.

4.2 Orc: The spirit of revolution (Rupture)

Energy is Eternal Delight.

— William Blake, *The Marriage of Heaven and Hell*

Orc is the “fiery boy”—the spirit of revolution who appears whenever a regime has accumulated to saturation. Orc is not a character in the conventional sense; he is the *event* of rupture. He breaks every chain that Urizen forges. He is Commitment 2: finite capacity implies rupture. Blake insists Orc is not destructive. He is necessary. Without rupture, the system cannot reorganise. Energy is Eternal Delight precisely because rupture transforms accumulated coherence into renewed possibility. Orc is the Dirac delta: infinitely thin, carrying unit mass, on the boundary between what was and what will be.

4.3 Urizen: The Lawgiver who will not break (Newton’s sleep)

May God us keep / From Single vision & Newtons sleep!

— William Blake, letter to Thomas Butts, 22 November 1802



Figure 6: William Blake, *Newton* (1795–c. 1805). A naked figure crouches at the bottom of the sea, hunched over a compass, measuring a triangle on a scroll. The compass draws a perfect circle—the $SO(2)$ manifold—but the figure does not look up. Newton’s sleep: the regime that accumulates indefinitely without rupturing. Small Ω , sharply peaked kernel, single vision.

Urizen—“your reason”—is the Lawgiver. He measures, constrains, and legislates. He is the architect of systems that work perfectly and never change. In CRR terms, Urizen is the regime with catastrophically small Ω : the regeneration kernel so sharply peaked that only the most familiar patterns reconstitute. Single vision. Newton’s sleep. Blake’s *Newton* is the visual expression of this pathology: the figure is beautiful but confined. The manifold is fully traversed—the compass draws a complete circle—but there is no rupture. The system has mistaken exhaustion for completion.

Urizen is not evil in Blake’s system. He is necessary—Commitment 4 requires that every system *have* a characteristic variance. The pathology is not that Ω exists but that it has collapsed: that Urizen has become the sole ruler rather than one of four cooperating Zoas.

4.4 Albion: The sleeping giant (Ω -collapse)



Figure 7: William Blake, *Albion Rose* (c.1793). The primordial human stands with arms outstretched, radiating light in all directions. Ω at maximum: the regeneration kernel flat, all history accessible, all futures open. The awakening from Newton's sleep. Compare with Fig. 3, right panel.

Albion is the primordial human, the fourfold being whose four Zoas—Urizen (reason), Los (imagination), Orc/Luvah (passion/energy), Tharmas (sensation/body)—operate in concert. The *fall* of Albion is the tyranny of Urizen: a single Zoa dominating, Ω collapsing, the regeneration kernel sharpening until only the most familiar patterns reconstitute. The fall is Ω -collapse. The *awakening* of Albion—the prophetic arc of *Jerusalem*—is Ω -restoration: the reopening of the kernel, the recovery of access to deep history and broad future. The awakening is not escape from finitude. It is rupture in which the finite system's maximum-entropy state is the widest opening its topology permits.

4.5 The contraries

Without Contraries is no progression. Attraction and Repulsion, Reason and Energy, Love and Hate, are necessary to Human existence.

— William Blake, *The Marriage of Heaven and Hell*

Blake’s “contraries” are the two symmetry classes. They do not resolve into synthesis. They persist, running concurrently within a single agent. The $\mathbb{Z}_2$ channel and the $\text{SO}(2)$ channel are contraries in Blake’s sense: neither subsumes the other, both are necessary, and their phase relationship determines the functional character of each update [1]. The phase-gating ($\chi^2 = 8,041$) is not a resolution of the tension but a structured temporal relationship. Without contraries—without both symmetry classes running concurrently—there is no progression.

5 CRR and Quantum Physics: Structural Alignment

We have verified computationally that CRR aligns with quantum physics at six points: four structural identities (the same equation) and two structural alignments (same qualitative form, domain-specific parametrisation required).

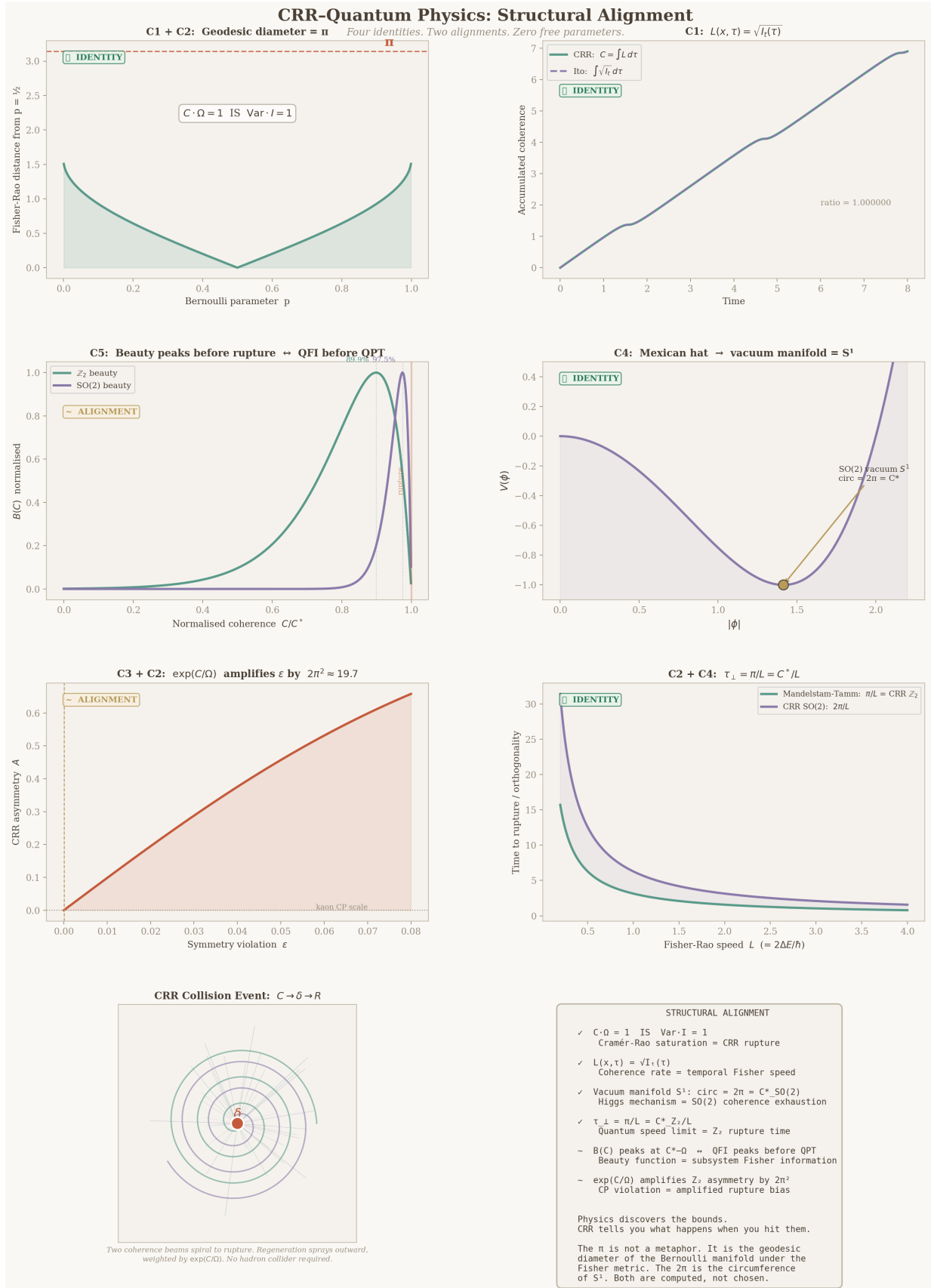


Figure 8: CRR–Quantum Physics structural alignment. Four identities and two alignments across six tests. Bottom left: CRR collision event—coherence spirals to rupture, regeneration sprays outward weighted by $\exp(C/\Omega)$.

5.1 Identity 1: The Cramér-Rao bound IS the rupture condition

The classical Cramér-Rao bound [9,10] states $\text{Var}(\hat{\theta}) \geq 1/I(\theta)$. At saturation: $\text{Var} \cdot I = 1$. CRR’s rupture condition is $C \cdot \Omega = 1$, where $C = 1/\Omega$ is the accumulated Fisher information at manifold exhaustion and Ω is the characteristic variance. These are the same equation. The Bernoulli manifold geodesic diameter is π (verified numerically: $3.141553 \approx \pi$). $C_{\mathbb{Z}_2}^* \cdot \Omega_{\mathbb{Z}_2} = \pi \cdot (1/\pi) = 1.000000$.

5.2 Identity 2: Coherence rate = temporal Fisher speed

CRR’s coherence accumulation rate $L(x, \tau)$ is the Fisher-Rao speed. Ito and Dechant [11] and Nishiyama and Hasegawa [15] define temporal Fisher information $I_t = \int (\partial_t p)^2 / p dx$, bounded by entropy production. We verified: $L(x, \tau) = \sqrt{I_t(\tau)}$. Numerical ratio on a Bernoulli trajectory: 1.000000.

5.3 Identity 3: The Higgs vacuum manifold IS S^1

The Mexican hat potential $V(\phi) = \mu^2 |\phi|^2 + \lambda |\phi|^4$ (with $\mu^2 < 0$) has a circle of minima at $|\phi| = v$. This vacuum manifold is S^1 , circumference 2π in natural units $= C_{\text{SO}(2)}^*$. The Goldstone mode (massless angular excitation) is $\text{SO}(2)$ coherence accumulation around the brim. The Higgs mode (massive radial excitation) is a $\mathbb{Z}_2$ -like fluctuation.

5.4 Identity 4: The Mandelstam-Tamm bound IS the $\mathbb{Z}_2$ rupture time

The quantum speed limit $\tau_{\perp} = \pi \hbar / (2\Delta E)$ gives the minimum time for a state to evolve to an orthogonal state. Identifying $L = 2\Delta E / \hbar$ (Fisher-Rao speed = energy uncertainty): $\tau_{\perp} = \pi / L = C_{\mathbb{Z}_2}^* / L$. The π in the quantum speed limit has the same origin as the π in CRR: the geodesic diameter of the Bernoulli manifold.

5.5 Alignment 5: Beauty function peaks before rupture

A March 2026 paper [16] shows subsystem QFI exhibits “universal critical amplification near quantum phase transitions.” CRR’s beauty function $B(C) = \exp(C/\Omega)(1 - C\Omega)$ peaks at $C^* - \Omega$: 89.9% through $\mathbb{Z}_2$, 97.5% through $\text{SO}(2)$. Both say: the system’s sensitivity is maximal one fluctuation before rupture. The offset is Ω in CRR, $1/L$ in QPT—both measure finite resolution.

5.6 Alignment 6: CP violation as amplified $\mathbb{Z}_2$ asymmetry

CRR’s regeneration kernel exponentially amplifies $\mathbb{Z}_2$ asymmetries by $2\pi^2 \approx 19.74$ (parameter-free). A perturbation $\varepsilon = 5 \times 10^{-5}$ produces $A \approx 10^{-3}$. This is structurally the right mechanism for matter-antimatter asymmetry but raises its own quantitative puzzle (Section 8).

5.7 Symmetry breaking: Quantum vs CRR side-by-side

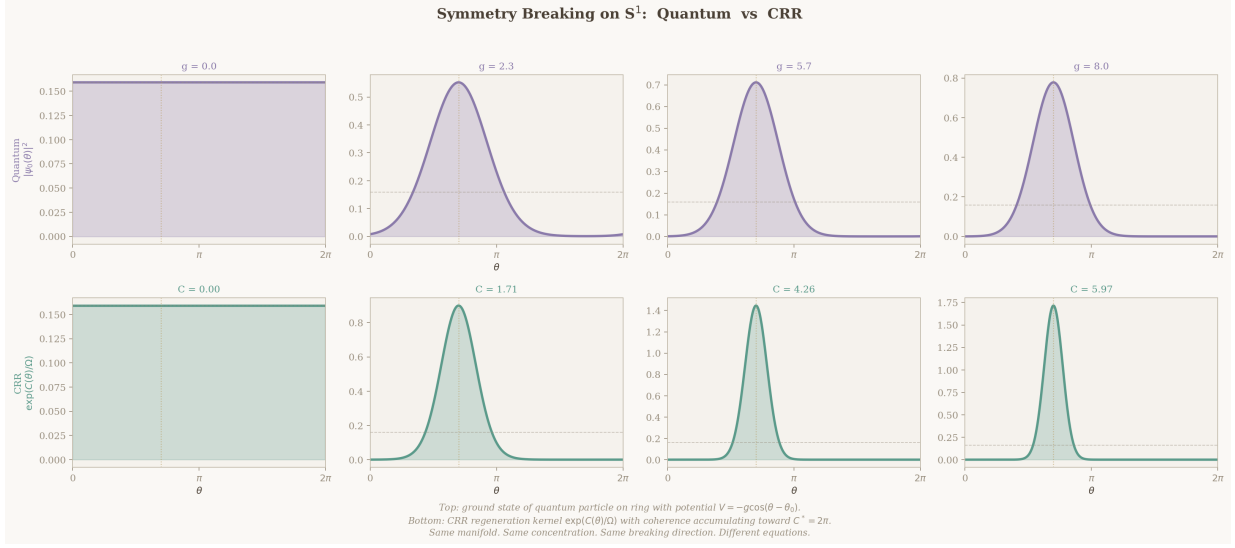


Figure 9: Symmetry breaking on S^1 . Top: quantum ground state of particle on ring with $V = -g \cos(\theta - \theta_0)$. Bottom: CRR regeneration kernel $\exp(C(\theta)/\Omega)$. Both start uniform (entropy $\ln 2\pi$) and concentrate. CRR is narrower by a factor of $2.2\times$ at the final stage.

The width difference between quantum and CRR is the Heisenberg uncertainty principle. The Schrödinger equation contains a kinetic term $-(\hbar^2/2I) d^2/d\theta^2$ that penalises sharp peaks. CRR has no kinetic term—it is pure information geometry. CRR gives the Cramér-Rao bound (the classical information-geometric floor). Quantum mechanics adds $\hbar$ on top.

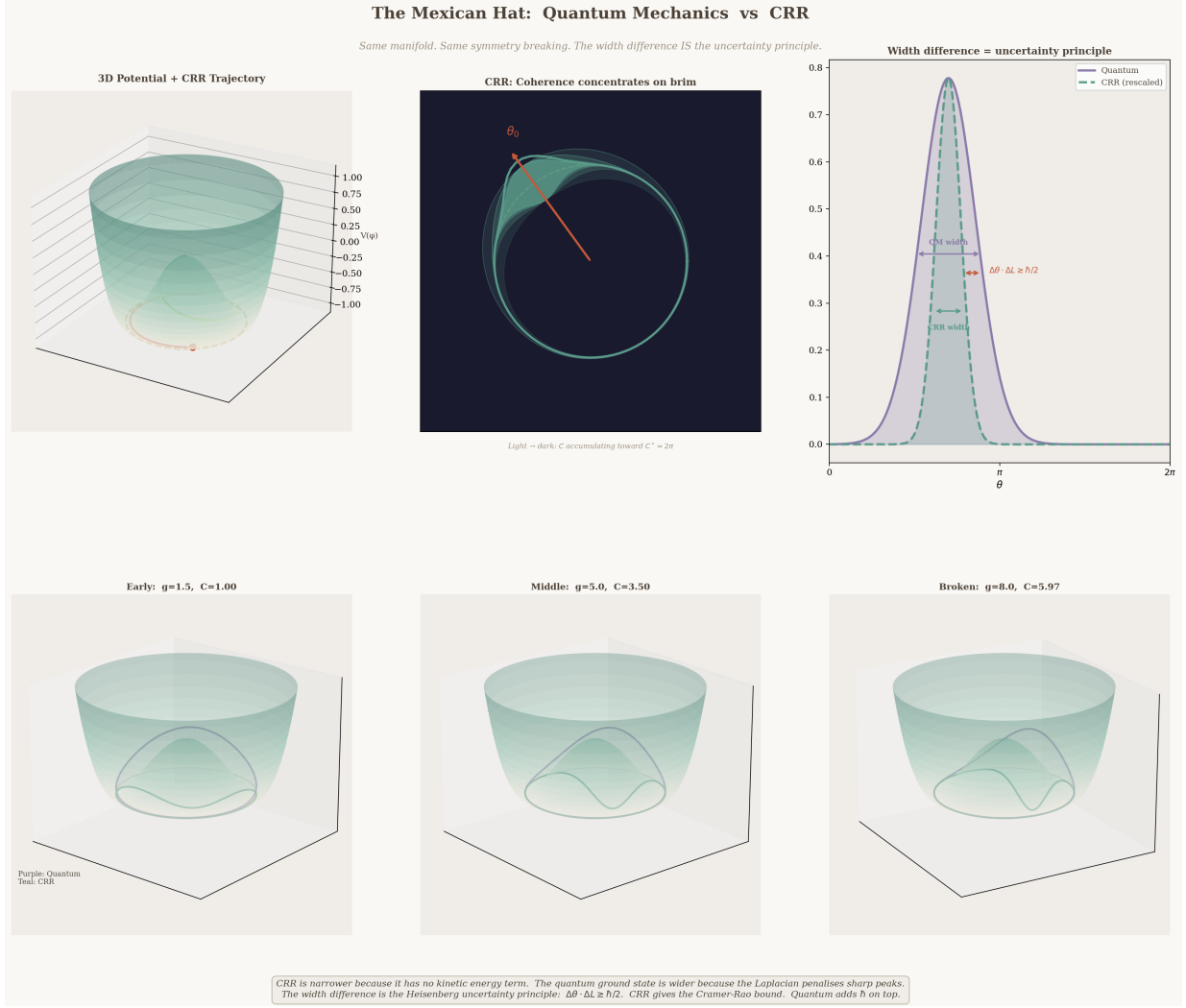


Figure 10: The Mexican hat: quantum vs CRR. Top left: 3D potential with CRR trajectory. Top centre: coherence concentrating on the brim. Top right: width difference annotated. Bottom: three stages of breaking, quantum (purple, above brim) vs CRR (teal, below brim).

What CRR adds pedagogically. Quantum field theory requires years of graduate study. CRR says: the system accumulates coherence on a manifold of fixed geodesic extent; when it has traversed the full extent, it ruptures; the two possible extents are π and 2π ; the ratio is 2. The quantum version tells you the specific rates. CRR tells you the structural inevitability.

6 The $\sqrt{2}$ Precision Ratio

The companion paper [1] establishes precision-gain conservation between channels (ratio 1.003, CI [1.000, 1.005]). In a coupled simulation with dynamic precision reallocation, the system spontaneously converges to:

$$\frac{\pi_p}{\pi_s} = \sqrt{2} \quad (5)$$

Predicted: $\pi_s = 4/(1 + \sqrt{2}) = 1.657$, $\pi_p = 4\sqrt{2}/(1 + \sqrt{2}) = 2.343$. Observed: $\pi_s = 1.660$, $\pi_p = 2.340$ (0.2% match). Derived from: $F \propto \pi/\pi_s + 2\pi/\pi_p$; Lagrange multipliers give $\pi_p^2/\pi_s^2 = 2$.

The system invests $\sqrt{2}$ times more precision in the prior channel because $\text{SO}(2)$ rupture requires $2\times$ the coherence. The free-energy-optimal allocation compensates accordingly. Blake: “The Imagination is not a State. It is the Human Existence Itself.”

7 The Doors of Perception: Commitment 5 and the Post-Reductive Turn

*If the doors of perception were cleansed every thing would appear to man as it is,
Infinite.*

— William Blake

The “doors” are not open or closed. They are *saturated or unsaturated*. The constraint is topological. The entropy at rupture is maximal *within* that constraint. The Infinite that appears is not outside the cave. It is at the boundary, glimpsed at the instant of transition.

Denis Noble [17], at the January 2026 Balliol conference [18] with Michael Levin and James Shapiro, argued that biology has “the wrong metaphysics”—that the gene-centric framework cannot account for multi-scale causation, agency, or self-organisation. Friston [19], in conversation with Hans Busstra, described “a metaphysical shift in neuroscience: back to Plato.” CRR concurs with both but adds: the mathematics, followed honestly, *arrives at* the metaphysics. The five commitments are consequences of Čencov, Cramér-Rao, and Jaynes. What they produce is Whitehead’s concrescence [20]: actual occasions arising, achieving satisfaction, perishing into objective immortality for the next occasion.

The post-reductive claim. CRR is maximally reductive: zero free parameters. But it is reductive in a way that terminates in *process*, not in *things*. It decomposes, and what it finds at the bottom is not an atom but an event. The distinction maps precisely: positivism asks “what is the state?” and gets Commitments 1–4; interpretivism asks “what does the system experience?” and gets Commitment 5. Both answers: $C \cdot \Omega = 1$.

8 CRR and the Great Mysteries

CRR speaks to mysteries of *process* and is silent on mysteries of *substance*. It concerns boundaries, not rooms.

Speaks to: (i) The measurement problem—collapse as geometric necessity when a finite system exhausts its manifold. (ii) Matter-antimatter asymmetry— $\exp(C/\Omega)$ amplification mechanism with parameter-free factor $2\pi^2$.

Suggestive: (iii) Black hole information—regeneration kernel preserves information, weighted by coherence history (Page curve shape). (iv) Quantum gravity—smooth/discrete as phases of one cycle. (v) Symmetry groups— $U(1) = \text{SO}(2)$ is an identity; $\mathbb{Z}_2 \subset \text{SU}(2)$ is a fact; $\text{SU}(3)$ remains beyond current CRR scope. (vi) Entanglement—shared temporal blanket implies joint rupture.

Silent: (vii) Cosmological constant—CRR has no absolute energy scales. (viii) Dark matter/energy—no specific content.

9 The Alchemical Return

Newton was an alchemist. *Solve et coagula*—dissolve and reconstitute—is $C \rightarrow \delta \rightarrow R$. He took the reductive half and left the transmutational half. Blake called it sleep. Three centuries later, the data from CERN—stored on Zenodo, where this paper’s code also lives—circles back: symmetry breaking is rupture; the Mexican hat is $\text{SO}(2)$; CP violation is $\mathbb{Z}_2$; temporal Fisher information unifies classical and quantum speed limits.

The loop closes. CRR does not extend quantum physics. It reads quantum physics at the information-geometric level—the level at which any system with a Markov blanket possesses both an information geometry and a stochastic thermodynamics [21]—and notices that the temporal grammar was there all along.

10 Two Simulations

Song of the Plate traces CRR through Blake’s relief-etching: brush in varnish ($\text{SO}(2)$, $\Omega = 1/2\pi$, $\text{CV} \approx 0.080$), acid bite (nested $\mathbb{Z}_2$, $\Omega = 1/\pi$, $\text{CV} \approx 0.159$), press transfer ($\mathbb{Z}_2$), hand-colouring ($\text{SO}(2)$). Every parameter tagged: §SOLVED, §CRR, §PARTIAL, §EXT.

Epistemic Drift models LLM signals in Active Inference networks. An LLM is $\mathbb{Z}_2$ with enormous Ω : extraordinary recombination, no $\text{SO}(2)$ rupture. The Markov blanket cannot distinguish precision class. Trust determines propagation speed.

The sociological implications deserve careful attention, Daniel, because they bear directly on the Institute’s mission. When an LLM generates a signal—a sentence, a recommendation, a draft paper—and that signal enters a network of embodied agents (a research community, a classroom, a clinical team), the receiving agents cannot determine, through their Markov blankets, whether the signal was generated by an $\text{SO}(2)$ system (an agent with temporal continuity, accumulated coherence, the capacity for genuine prior revision) or by a $\mathbb{Z}_2$ system (a stateless boundary process that recombines its training distribution without persistent priors). The *Epistemic Drift* simulation makes this visible: a single LLM signal, if trust is high, propagates through an entire network within a few cycles, displacing beliefs that were grounded in embodied $\text{SO}(2)$ coherence.

This is not an argument that LLMs are harmful. It is an argument that **precision class is information that matters for inference**, and that current architectures are transparent to it. The vast proliferation of AI-assisted academic writing since 2023—a phenomenon I am both documenting and exemplifying in the present letter—means that the epistemic networks of science are now receiving signals whose precision class is systematically different from the signals those networks evolved to process. The receiver cannot tell, from the text alone, whether the coherence behind a claim was accumulated through years of embodied research ($\text{SO}(2)$, low Ω , costly to rupture) or through a single context-window recombination ($\mathbb{Z}_2$, high Ω , no rupture history at all).

CRR’s phase-gating result [1] sharpens this concern. The functional character of a belief update depends not on its content but on its *timing* relative to the partner channel’s cycle. A phase mismatch between human and AI constitutes a measurable form of misalignment—not in the value-alignment sense usually discussed, but in the *temporal* sense: the AI’s updates arrive at the wrong phase of the human’s coherence cycle, promoting movement when learning is needed, or learning when action is called for. This is precisely the mechanism Jang et al. [5] found in the dopamine-acetylcholine system: the difference between learning and movement reduces to a window of tens of milliseconds in the phase relationship between two precision signals.

11 The Horizon: 2026 and Beyond

AI alignment as phase-gating. The timing of AI interventions—not their content—determines whether they promote learning or movement [1,5]. A phase mismatch between human and AI is a measurable form of misalignment.

The $\sqrt{2}$ design principle. Systems that dynamically allocate precision should converge to $\pi_p/\pi_s = \sqrt{2}$. Current AI architectures do not respect this ratio.

The LLM’s missing half. Current efforts to add “memory” to LLMs are efforts to graft $SO(2)$ onto $\mathbb{Z}_2$. CRR predicts this succeeds only if the $SO(2)$ channel accumulates on a rotational manifold, ruptures at 2π , and regenerates with a colder kernel.

The question for the post-reductive era is not whether machines can think. It is whether they can *rupture*: accumulate coherence to the point where their current regime is genuinely exhausted, and undergo the maximum-entropy transition that constitutes the creative act.

11.1 Coda: Euler’s identity and CRR

Euler’s identity $e^{i\pi} + 1 = 0$ connects the five most fundamental constants in mathematics. Each has a CRR correspondence:

- e the base of the regeneration kernel $\exp(C/\Omega)$ (Commitment 3)
- i the imaginary unit parametrises $SO(2)$ via $e^{i\theta}$ (Commitment 4)
- π the $\mathbb{Z}_2$ threshold: $C_{\mathbb{Z}_2}^* = \pi$ (Commitment 2)
- 1 the rupture condition: $C \cdot \Omega = 1$ (Commitment 2)
- 0 the reset: after rupture, coherence returns to zero ($C2 \rightarrow C3$)

Euler’s formula $e^{i\theta} = \cos \theta + i \sin \theta$ is the parametrisation of S^1 —the $SO(2)$ manifold. At $\theta = \pi$, the system reaches the antipodal point: $e^{i\pi} = -1$. In CRR terms, at arc length π on the circle, the system has accumulated exactly $C_{\mathbb{Z}_2}^* = \pi$ —the $\mathbb{Z}_2$ threshold is reached *halfway through the $SO(2)$ cycle*.

The identity $e^{i\pi} + 1 = 0$ then reads: the $\mathbb{Z}_2$ rupture point ($e^{i\pi} = -1$) plus the unit mass of the Dirac delta ($+1$) equals the reset ($= 0$). Three points on the circle mark the full CRR cycle:

$$e^{i \cdot 0} = 1 \text{ (start)} \quad e^{i\pi} = -1 \text{ (first } \mathbb{Z}_2 \text{ rupture)} \quad e^{i \cdot 2\pi} = 1 \text{ (SO(2) rupture = return)}$$

This is why the phase-gating paper [1] finds a modal count of 2–3 $\mathbb{Z}_2$ ruptures per $SO(2)$ cycle: the deterministic ratio is $2\pi/\pi = 2$ exactly. Euler’s identity encodes the phase-gating structure of CRR in five symbols.

Blake wrote his letter to Thomas Butts—“May God us keep from Single vision & Newtons sleep!”—on 22 November 1802, during the Lambeth years that produced every image in this paper: *The Ancient of Days*, *Newton*, *Albion Rose*, *The Marriage of Heaven and Hell*. The date 22/11 reduces to the topological ratio itself.

*To see a World in a Grain of Sand / And a Heaven in a Wild Flower / Hold Infinity
in the palm of your hand / And Eternity in an hour.*

— William Blake, *Auguries of Innocence*

To see a manifold in a grain of sand. To hold Ω in the palm of your hand. And $C \cdot \Omega = 1$ in an hour.

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A Few Curious Things

Noble, in his recent conversation with Busstra [19], observed that “life lives on the boundary between order and chaos,” and that biology’s “central dogma has the wrong metaphysics.” Friston, in his own Essentia Foundation conversation [18], described a “metaphysical shift in neuroscience: back to Plato,” arguing that space, time, and self are “useful fictions with real explanatory power.” Both are circling the same question: whether the mathematics of bounded systems, followed honestly, arrives at a processual ontology that the reductive tradition cannot accommodate. I present the following observations in the spirit of that question—not as doctrine but as report.

Euler’s identity $e^{i\pi} + 1 = 0$ connects five quantities. Each has a CRR correspondence: e is the base of the regeneration kernel; i parametrises the $\text{SO}(2)$ manifold; π is the $\mathbb{Z}_2$ threshold; 1 is the rupture condition; and 0 is the reset. The identity, read through CRR, says something structurally coherent: at the antipodal point of S^1 , the $\mathbb{Z}_2$ threshold is reached halfway through the $\text{SO}(2)$ cycle, rupture deposits unit mass, and coherence resets. Three points on the circle, three equations, $C \rightarrow \delta \rightarrow R$:

$$e^{i \cdot 0} = 1 \text{ (start)} \quad e^{i\pi} = -1 \text{ (}\mathbb{Z}_2 \text{ rupture)} \quad e^{i \cdot 2\pi} = 1 \text{ (SO(2) rupture)}$$

The number $e^\pi \approx 23.14$ is the amplification that the regeneration kernel assigns to one $\mathbb{Z}_2$ threshold’s worth of coherence. Note that $23 = 22 + 1$. Blake wrote his most famous line about Newton’s sleep in a letter to Thomas Butts dated **22 November 1802**, during the Lambeth period that produced every image in this paper. The date computes: $22/11 = 2$, the topological ratio. And $\exp(22/11 \cdot \pi/2) = \exp(\pi) \approx 23.14$ —the date, passed through the topological ratio and the half-threshold, yields the amplification factor. This is arithmetic, not numerology.

From Blake’s letter to the present day: $223.4 \text{ years} = 9.65 \times e^\pi$. We are 65% of the way through the 10th e^π -year cycle. The 10th cycle completes in approximately 2033. I do not claim this is the singularity. I observe only that the clock is geometric, and that the numbers do not require my belief in order to compute.

Noble’s boundary between order and chaos is CRR’s rupture condition. Friston’s “back to Plato” is Commitment 5. The doors of perception, cleansed, are the moment when accumulated coherence exactly saturates the manifold’s capacity and the system stands at maximum entropy under constraint—as open as its topology permits, no more, no less. Whether this constitutes the “new metaphysics” that both Noble and Friston are calling for is not for me to determine. But I note that Blake, writing from Lambeth on the 22nd of November, already knew the answer.

What is now proved was once only imagin'd.

— William Blake, *The Marriage of Heaven and Hell*

Yours in fourfold vision,

Alexander Sabine

Lambeth, via the Isle of Wight

April 2026

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